

Phase 02 - Week 06 - Day 02 - 06Sep2026 01

Resume RAG Enhancement — Assignment Report

****Project:**** RecruitBot — Enhanced Resume RAG Application

1. Assignment

Enhance the existing Resume RAG application by implementing:

- Re-ranking of retrieved resumes
- Deduplication of repeated results
- Summarization of the final results
- Complete end-to-end pipeline integration
- Display all relevant results clearly on the UI

And customize with organization branding: name and logo, brand theme and colours, professional UI layout, responsive design, and loading / success / validation / error messages. Then test the complete end-to-end application.

2. Solution Overview

RecruitBot is a recruiter-focused web application. A recruiter describes a role in plain language and receives ****ranked, deduplicated, AI-summarized**** candidate matches from a resume database.

****Two applications:****

App	Tech	Port	Role
Backend (`resume-rag-backend`)	Node + Express + TypeScript, MongoDB Atlas, Mistral (embeddings), Groq (LLM)	3000	Retrieval pipeline + ingestion API
Frontend (`recruitbot-web`)	Vite + React 18 + TypeScript + TailwindCSS + Zustand	5173	Search chat UI + upload UI

****Enhanced retrieval pipeline**** — one call to `POST /v1/search` runs:

...

Query

→ Embed (Mistral)

- BM25 search + Vector search (in parallel)
- Merge + Deduplicate (by resume id)
- Re-rank with LLM (Groq) — relevance score + reason
- Summarize each result (Groq) — fit summary
- Ranked, deduplicated, summarized results

...

The database currently holds **168 resumes**, all embedded and searchable.

3. Features Implemented

3.1 Re-ranking of retrieved resumes

In **Hybrid** mode, after retrieval an LLM re-orders candidates by how well they fit the query, assigning each a **relevance score** and a short **"why"** reason. Results are shown in ranked order (#1, #2, ...) with a coloured score pill and the reason on each card. Re-ranking is **always on** in Hybrid (not a toggle). Vector/BM25 modes show each engine's own ranking.

3.2 Deduplication of repeated results

Keyword (BM25) and semantic (Vector) searches often return the same candidate. These are **merged into single results** keyed by resume id. The UI shows this explicitly:

- A results-header line: **"Deduplicated — N candidates matched by both keyword + semantic search (merged)."**
- A per-card **"Keyword + Semantic"** badge.

3.3 Summarization of final results

Each top candidate gets an AI **"Fit Summary"** tailored to the query (shown truncated on the card, in full in the profile modal).

3.4 Complete end-to-end pipeline integration

A single **Hybrid** search triggers the whole chain (embed → BM25 + vector → merge → dedupe → re-rank → summarize) in one request, with per-stage timings and graceful degradation if the LLM is briefly unavailable.

3.5 Display all relevant results clearly on the UI

- Results header: "Found N candidates · Mode · time"
- Ranked candidate cards: rank badge, score pill, role, experience, dedup badge, skills, "why" reason, fit summary
- A candidate **profile modal** (opens on card click; closes via ✕, overlay click, or Esc)
- Graceful **empty state** and honest low-relevance results for off-topic queries

4. Organization Branding & UX Customization

Aspect	Choice
Name	**RecruitBot**
Logo	**"RB"** monogram in a blue→cyan gradient (also the favicon)
Theme	**Professional Blue**, dark — primary `#2563eb`, accent `#06b6d4`
Backgrounds	Cool blue-gray — `#0b1120` / `#111a2e` / `#172136`
Score colours	Vector `#3b82f6`, BM25 `#22d3ee`, Hybrid `#34d399`
Layout	Sidebar (brand, search modes, Hybrid-only AI panel, results limit) + chat main + candidate modal
Responsive	Mobile (<768px) collapses the sidebar into a slide-out drawer with a hamburger toggle
Messages	Loading, success, validation, and error states across search + upload

Search modes (each runs its true engine):

- **Vector** — semantic-only search (`/v1/search/vector`), ordered by similarity
- **BM25** — keyword-only search (`/v1/search/bm25`), ordered by BM25 relevance
- **Hybrid** — full AI pipeline (`/v1/search`): both engines → merge + dedupe → LLM re-rank → optional summaries, with adjustable BM25/Vector weights

AI Enhancements (shown only in Hybrid mode, where they apply):

- **Re-ranking** — always on (shown as an "AI-ranked" indicator, not a toggle)
- **Deduplication** — always on (shown as a "Deduplicated — N merged" indicator)
- **Summarize** — a user toggle (the heaviest LLM step, so it stays optional)

Vector and BM25 are pure single-engine searches (fast, no LLM), so they show each engine's own relevance order and do not display re-rank/dedup/summaries.

5. Messaging (loading / success / validation / error)

Type	Where	Example
Loading	Search	"Searching & ranking candidates..." with animated dots
Loading	Upload	Progress bar → pipeline stepper (upload → extract → parse → embed → store)
Success	Upload	Checklist screen + parsed summary
Success	Search	"Found N candidates · Mode · time" header
Validation	Upload	"Only PDF allowed", "Maximum 5MB allowed", "Please select a file"

Error	Search	Friendly, retryable card (timeout / can't reach server / unavailable / server error)
Error	Upload	Backend error codes mapped to friendly copy with retry
Degradation	Search	Amber notice when AI re-rank/summary is temporarily unavailable

6. End-to-End Testing

`npm run build` passes (TypeScript + Vite) with no errors. The full pipeline was verified live against the running backend. Each feature was tested one-by-one in the UI and confirmed:

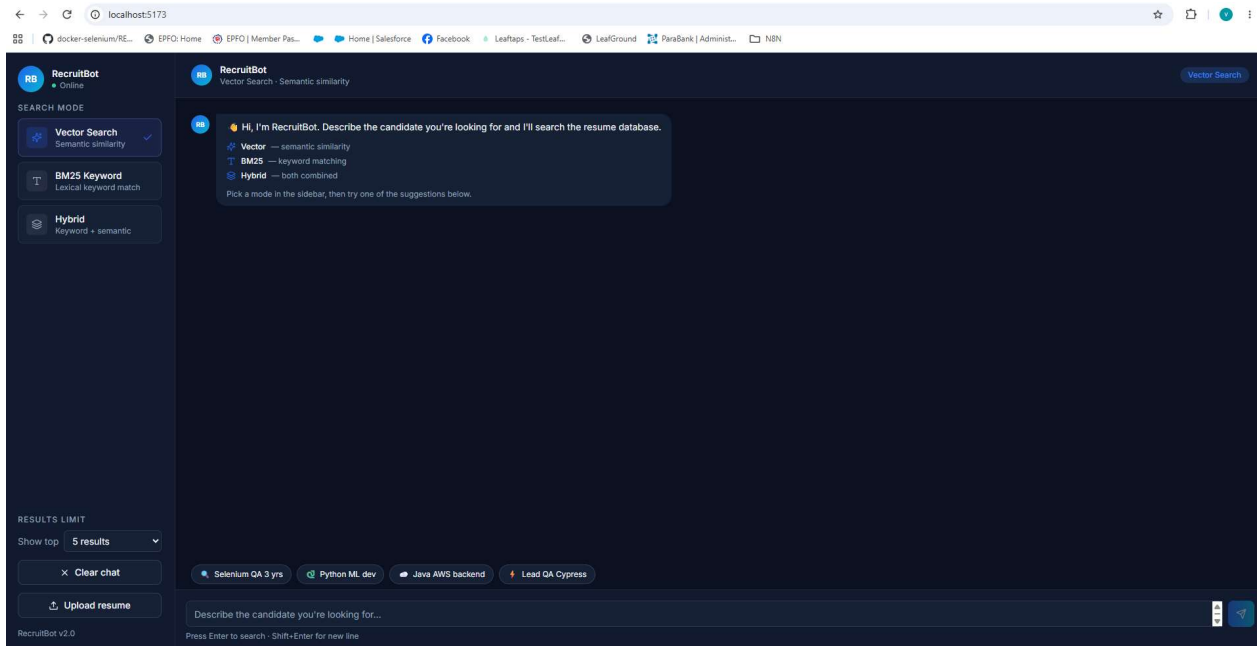
#	Feature	How tested	Result
1	Re-ranking	Query returned ranked #1–#5 with descending scores (0.950 → 0.850) + "why" reasons	✓
2	Deduplication	Hybrid query showed "Deduplicated — N merged" line + "Keyword + Semantic" badges, no duplicate cards	✓
3	Summarization	"FIT SUMMARY" boxes on cards + full summary in modal; toggling Summarize off removed them	✓
4	End-to-end	One search produced header + dedup line + scores + reasons + summaries together	✓
5	Display clarity	Clean layout + modal (open/close via ×, overlay, Esc); off-topic query returned honest low scores	✓

Typical successful run timings: embedding ~0.3–1s, BM25 ~0.15s, vector ~0.1s, re-rank ~2.5–3.7s, summarize ~0.9–2s, total ~4–6s.

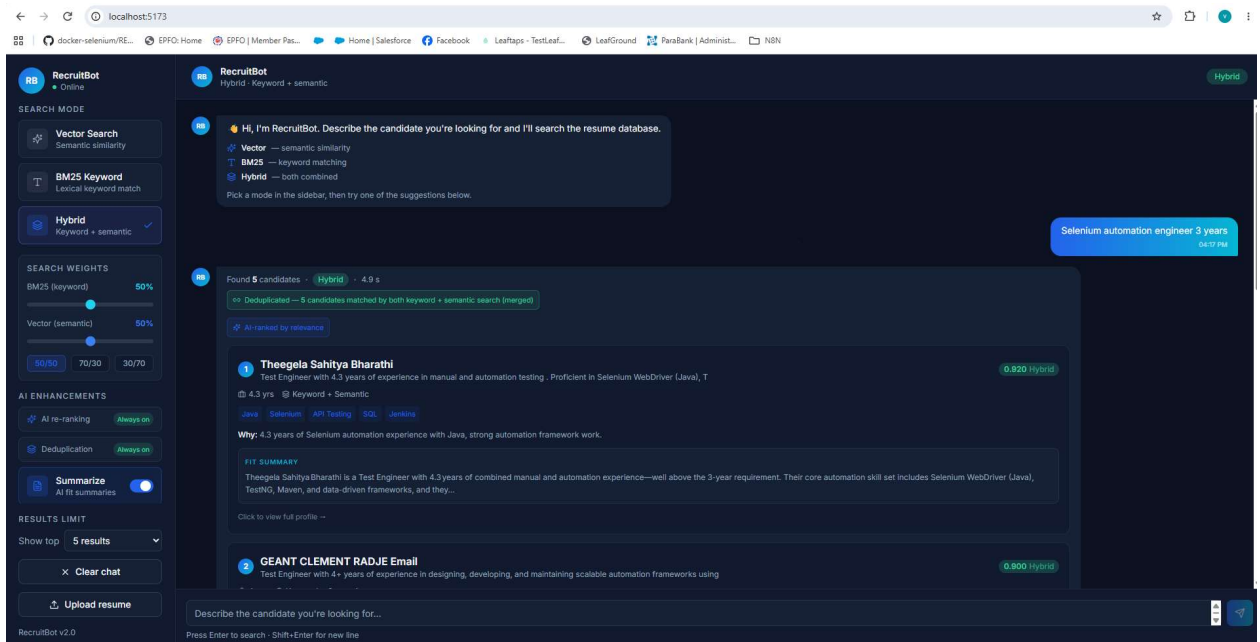
7. Screenshots

> Insert your screenshots below.

1. **Home / welcome screen**



2. **Re-ranking** — ranked cards with scores + reasons



3. **Deduplication** — "Deduplicated — N merged" line + badges

RecruitBot

Online

SEARCH MODE

Vector Search

Semantic similarity

BM25 Keyword

Lexical keyword match

Hybrid

Keyword + semantic

SEARCH WEIGHTS

BM25 (keyword)

50%

Vector (semantic)

50%

50/50

70/30

30/70

AI ENHANCEMENTS

AI re-ranking

Always on

Deduplication

Always on

Summarize

AI fit summaries

RESULTS LIMIT

Show top

5 results

Clear chat

Upload resume

RecruitBot v2.0

RecruitBot

Hybrid · Keyword + semantic

Hybrid

Hi, I'm RecruitBot. Describe the candidate you're looking for and I'll search the resume database.

Vector

Semantic similarity

BM25

Keyword matching

Hybrid

Both combined

Pick a mode in the sidebar, then try one of the suggestions below.

Selenium automation engineer 3 years

04/17 PM

Found 5 candidates · Hybrid · 4.9 s

Deduplicated — 5 candidates matched by both keyword + semantic search (merged)

AI-ranked by relevance

1

Theegela Sahitya Bharathi

Test Engineer with 4.3 years of experience in manual and automation testing . Proficient in Selenium WebDriver (Java), T

0.920 Hybrid

4.3 yrs

Keyword + Semantic

Java · Selenium · API Testing · Git · Jenkins

Why:

4.3 years of Selenium automation experience with Java, strong automation framework work.

FIT SUMMARY

Theegela Sahitya Bharathi is a Test Engineer with 4.3 years of combined manual and automation experience—well above the 3-year requirement. Their core automation skill set includes Selenium WebDriver (Java), TestNG, Maven, and data-driven frameworks, and they...

Click to view full profile →

2

GEANT CLEMENT RADJE Email

Test Engineer with 4+ years of experience in designing, developing, and maintaining scalable automation frameworks using

0.900 Hybrid

Describe the candidate you're looking for...

Press Enter to search · Shift+Enter for new line

4. ****Summarization**** — card with FIT SUMMARY box

Theegela Sahitya Bharathi

Test Engineer with 4.3 years of experience in manual and automation testing . Pr...

☆ 0.920 Hybrid

🏠 4.3 yrs experience

Matched by keyword + semantic search (merged)

SKILLS

Java

Selenium

API Testing

SQL

Jenkins

WHY THIS CANDIDATE

4.3 years of Selenium automation experience with Java, strong automation framework work.

FIT SUMMARY

Theegela Sahitya Bharathi is a Test Engineer with 4.3 years of combined manual and automation experience—well above the 3-year requirement. Their core automation skill set includes Selenium WebDriver (Java), TestNG, Maven, and data-driven frameworks, and they have a proven record of reducing manual effort by 45 % at Accenture. Additional relevant competencies are CI/CD integration with Jenkins, API

5. ****Candidate modal**** — score + matched-by + skills + why + full fit summary

Theegela Sahitya Bharathi

Test Engineer with 4.3 years of experience in manual and automation testing . Pr...

☆ 0.920 Hybrid

🏠 4.3 yrs experience

Matched by keyword + semantic search (merged)

SKILLS

JavaSeleniumAPI TestingSQLJenkins

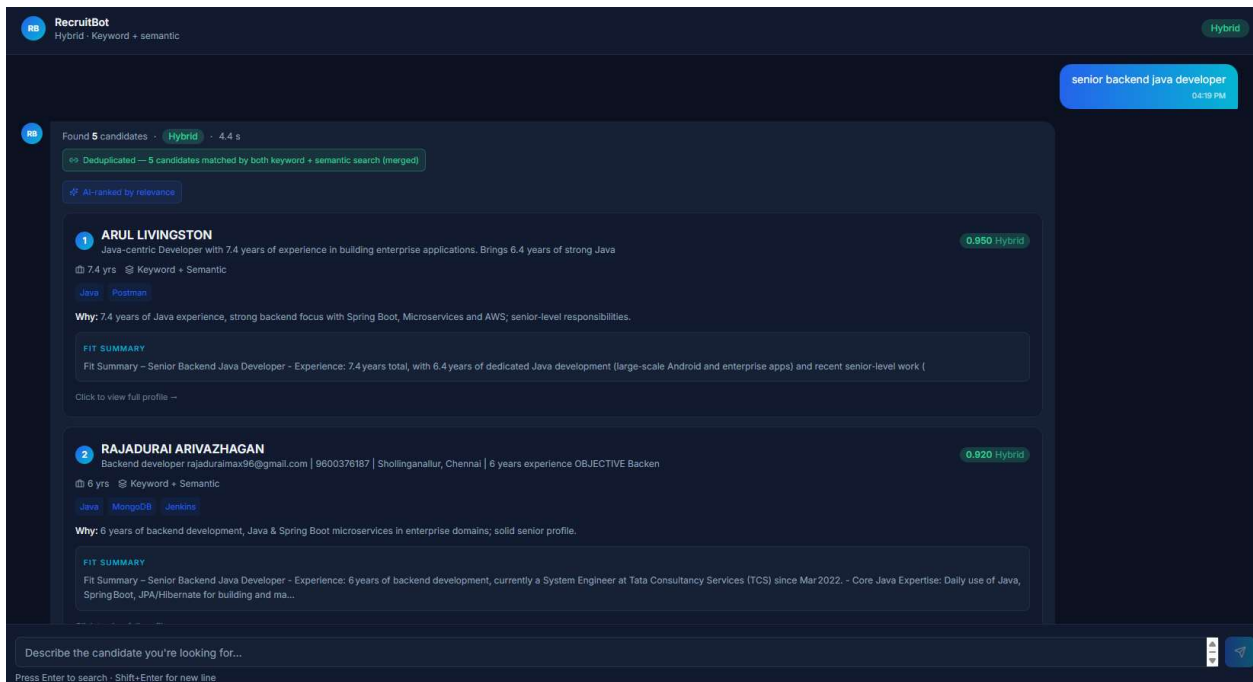
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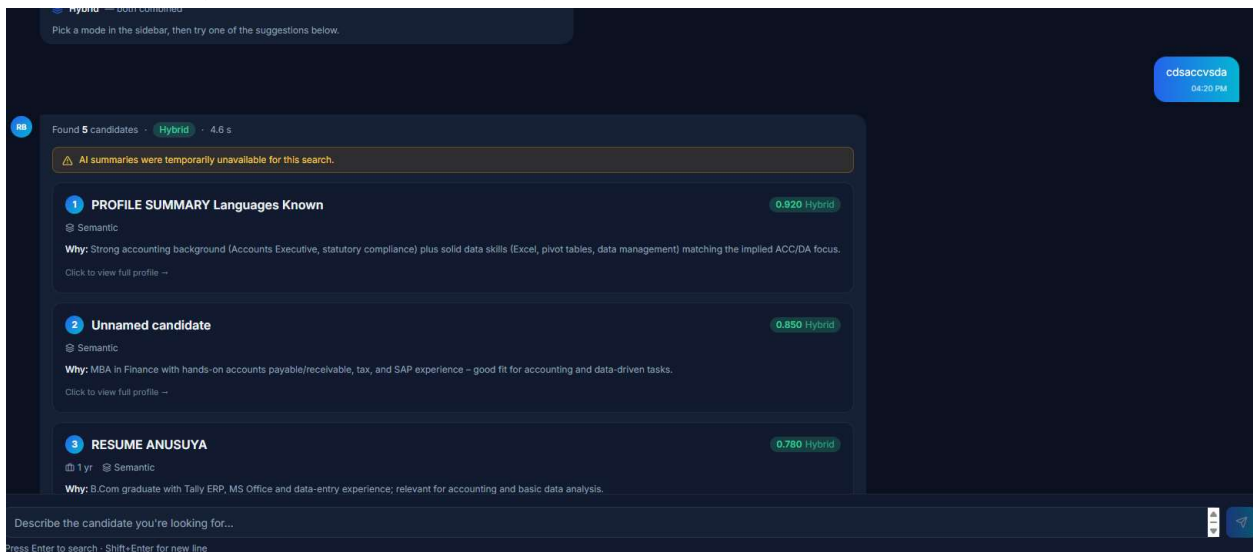
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6. **End-to-end** — one search showing everything together



7. **Empty / edge case** — off-topic query with honest low scores



8. **Resume upload** (`/ingest`)



RecruitBot
Resume Ingestion

Upload a resume

Add a candidate PDF to make it searchable in RecruitBot.



Ingestion complete

Abish - Marketing (1) (1).pdf



Resume uploaded successfully



Embedding generated (1024-dim · mistral-embed)



MongoDB ingestion completed



Vector search ready

ROLE

Marketing Specialist with pro...

EXPERIENCE

1.5 yrs

SKILLS PARSED

0

RESUME ID

6aa28b957bad5bde85363ff4

PROCESSED IN

2455 ms



Upload another

8. How to Run

The backend needs a `.env` with a MongoDB Atlas URI, a Mistral API key, and a Groq API key. (Never commit `.env` — it is git-ignored.)

```
``bash
# Backend → http://localhost:3000
cd resume-rag-backend
npm install
npm run dev
```

```
# Frontend → http://localhost:5173
cd recruitbot-web
npm install
npm run dev
``
```

- Candidate search: `http://localhost:5173/`
- Resume upload: `http://localhost:5173/ingest`

9. Repository

Supporting documents in the repo:

- `IMPLEMENTATION.md` — phase-wise implementation detail
- `PROMPTS.md` — prompt file
- `SUBMISSION.md` — submission checklist
- `README.md` — project overview

10. Honest Notes

- The Groq LLM (free tier) can rate-limit intermittently. The app degrades gracefully (shows a notice) and works on retry after a short wait.
- Some candidate names appear as extracted text fragments due to backend resume parsing on messy PDFs — a data/parsing matter, not a UI issue.
- The same person uploaded twice appears as two resume ids; deduplication merges per document (same document is merged correctly).
- The candidate modal is populated from the search result (the backend does not expose a separate candidate-detail endpoint).